# Technical Report: Fast-Growing Firms Prediction

Data Analysis 3 - Assignment 2
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## 1 Introduction

This technical report details the methodology, modeling choices, and evaluation results for predicting fast-growing firms using the Bisnode panel data (2010-2015). The code and full notebooks are available in the associated GitHub repository: `https://github.com/zarizachow/Data-Analysis-3`.

## 2 Data Preparation & Feature Engineering

### 2.1 Sample Design

We utilized the `bisnode-firms` dataset. The cleaning process involved:

- **Panel Construction:** We constructed a panel for 2010-2015.
- **Target Window:** The prediction target is defined over the 2012-2013 period.
- **Filtering:** We filtered for firms that were "alive" (positive sales) and had sales between 1,000 and 10 million EUR in the baseline year (2012).

### 2.2 Target Variable

The target `fast_growth` is a binary variable representing the top decile of sales growth.

$$\text{Growth} = \ln(\text{Sales}_{2013}) - \ln(\text{Sales}_{2012})$$

Firms with growth $\geq 90^{th}$ percentile were labeled as `1`, others `0`. The base rate of the positive class is approx. 10%.

### 2.3 Features

We engineered a rich set of features based on financial reports and firm characteristics:

- **Financials:** Log sales, assets, liabilities, profit/loss ratios.
- **Growth:** Lagged sales growth (`d1_sales_mil_log`).
- **Demographics:** Firm age, CEO age, region, industry category (`ind2_cat`).
- **Quality Flags:** Flags for data errors or extreme values in accounting fields.

Missing numeric values were imputed with the mean, and categorical values with the mode. One-hot encoding was applied to categorical variables.

The relationship between firm size and growth is notably non-linear, as seen below. This supports the use of tree-based models like Random Forest.

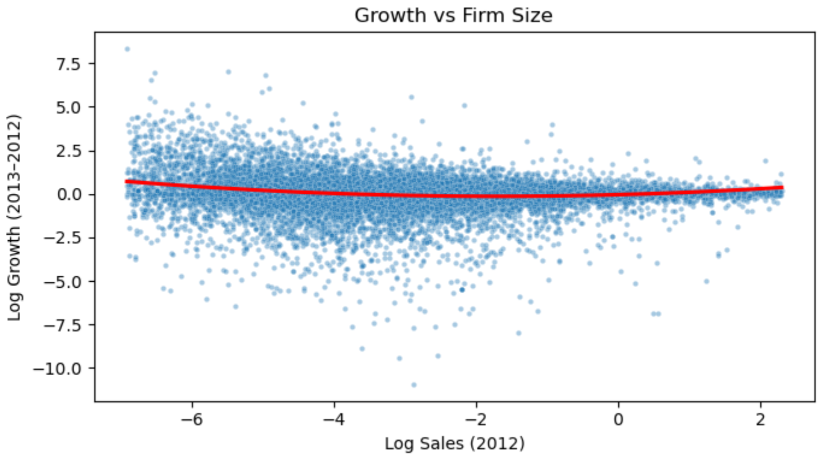


Figure 1: Growth vs. Firm Size (Log Sales)

# 3 Modeling Strategy

## 3.1 Models

We trained three models to predict the probability of fast growth:

1. **Logistic Regression (L2):** Standard parametric baseline.
2. **Logistic Regression (LASSO/L1):** For feature selection in high dimensions.
3. **Random Forest:** Non-parametric ensemble to capture non-linearities (e.g., Age vs. Growth).

## 3.2 Evaluation

We used 5-fold Stratified Cross-Validation.

- **Part I (Probability):** Metrics included ROC-AUC, Average Precision, and Brier Score.
- **Part II (Classification):** We defined a loss function with $Cost(FN) = 4 \times Cost(FP)$.

# 4 Results

## 4.1 Part I: Probability Prediction

The Random Forest model significantly outperformed the linear models.

| Model | ROC-AUC | Avg Precision | Log Loss | Runtime (s) |
|---|---|---|---|---|
| **Random Forest** | **0.828** | **0.460** | **0.275** | $\sim 3.5$ |
| Logit (LASSO) | 0.753 | 0.269 | 0.433 | $\sim 742$ |
| Logit (L2) | 0.753 | 0.269 | 0.433 | $\sim 50$ |

Table 1: Cross-validated performance metrics.

The Random Forest's superior performance (AUC 0.83 vs 0.75) highlights the importance of non-linear interactions in predicting firm growth.

### 4.2 Part II: Classification & Loss Minimization

We optimized the classification threshold $t$ to minimize the expected loss:

$$\text{Expected Loss} = \frac{1}{N}\sum(FP \times 1 + FN \times 4)$$

- **Random Forest:** Minimized Loss = **0.255** at Threshold ≈ **0.15**.
- **Logit (L2):** Minimized Loss = 0.316 at Threshold ≈ 0.45.

The lower threshold for Random Forest reflects the high cost of false negatives; the model is calibrated to cast a wider net to catch the valuable fast-growers.

### 4.3 Part III: Confusion Matrix

Below is the confusion matrix for the Random Forest model (Fold 0) at the optimal threshold.

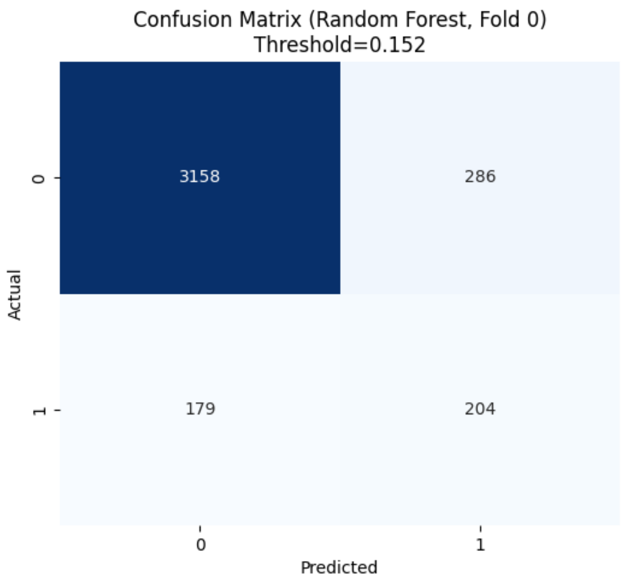


Figure 2: Confusion Matrix (Random Forest)

### 4.4 Part IV: Industry Analysis

We repeated the classification exercise for two subgroups:

1. **Manufacturing:** AUC = 0.817, Expected Loss = 0.239
2. **Services:** AUC = 0.838, Expected Loss = 0.257

The model maintains high performance across both sectors, validating its generalizability.

## 5 Code Snippet: Model Training

```
# Example pipeline construction
preprocess = ColumnTransformer(
    transformers=[
```

```python
        ("num", numeric_transformer, num_cols),
        ("cat", categorical_transformer, cat_cols),
    ]
)

rf_model = RandomForestClassifier(
    n_estimators=300,
    min_samples_leaf=10,
    random_state=42,
    n_jobs=-1
)

rf_pipe = Pipeline(steps=[("prep", preprocess), ("model", rf_model)])

# Cross-validation loop for loss minimization
for train_idx, test_idx in cv.split(X, y):
    rf_pipe.fit(X.iloc[train_idx], y.iloc[train_idx])
    probs = rf_pipe.predict_proba(X.iloc[test_idx])[:, 1]
    # ... threshold calculation ...
```

## 6 Conclusion

The Random Forest model is the clear choice for this prediction task. Its ability to model complex, non-linear relationships results in superior ranking capability (high AUC) and lower business risk (low Expected Loss).